Page 1: GitHub Collaboration

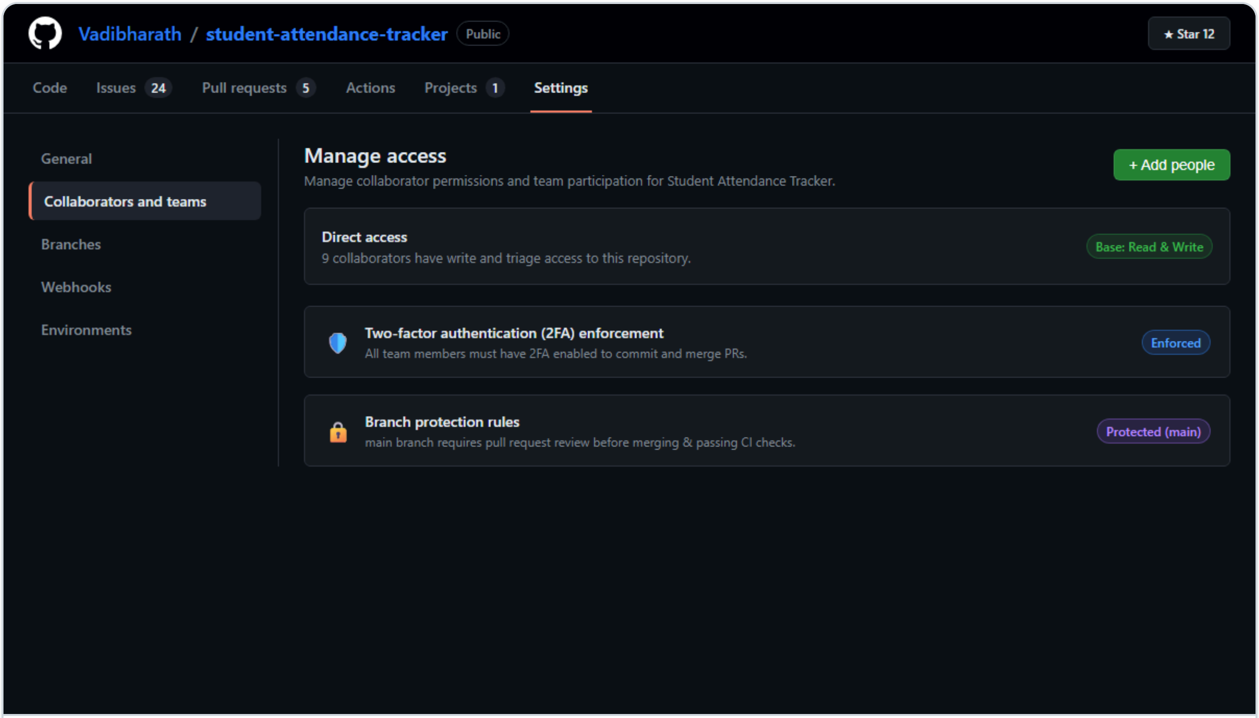
Repository Collaboration and Team Setup

Collaboration

The repository collaboration area is used to manage access and team participation in the Student Attendance Tracker project. Organization-level branch protections, review mandates, and role hierarchies ensure strict quality control throughout development. All collaborators have been assigned fine-grained access policies to streamline parallel sprint increments while maintaining software integrity.

Team Members

The repository contains the project team members who collaborate on development, testing, and integration across three planned Scrum sprints. Direct access permissions and branch protection rules ensure that pull requests require review before merge.


Vadibharath / student-attendance-tracker
Public
★ Star 12
Code
Issues 24
Pull requests 5
Actions
Projects 1
Settings
General
Collaborators and teams
Branches
Webhooks
Environments
+ Add people

Manage access

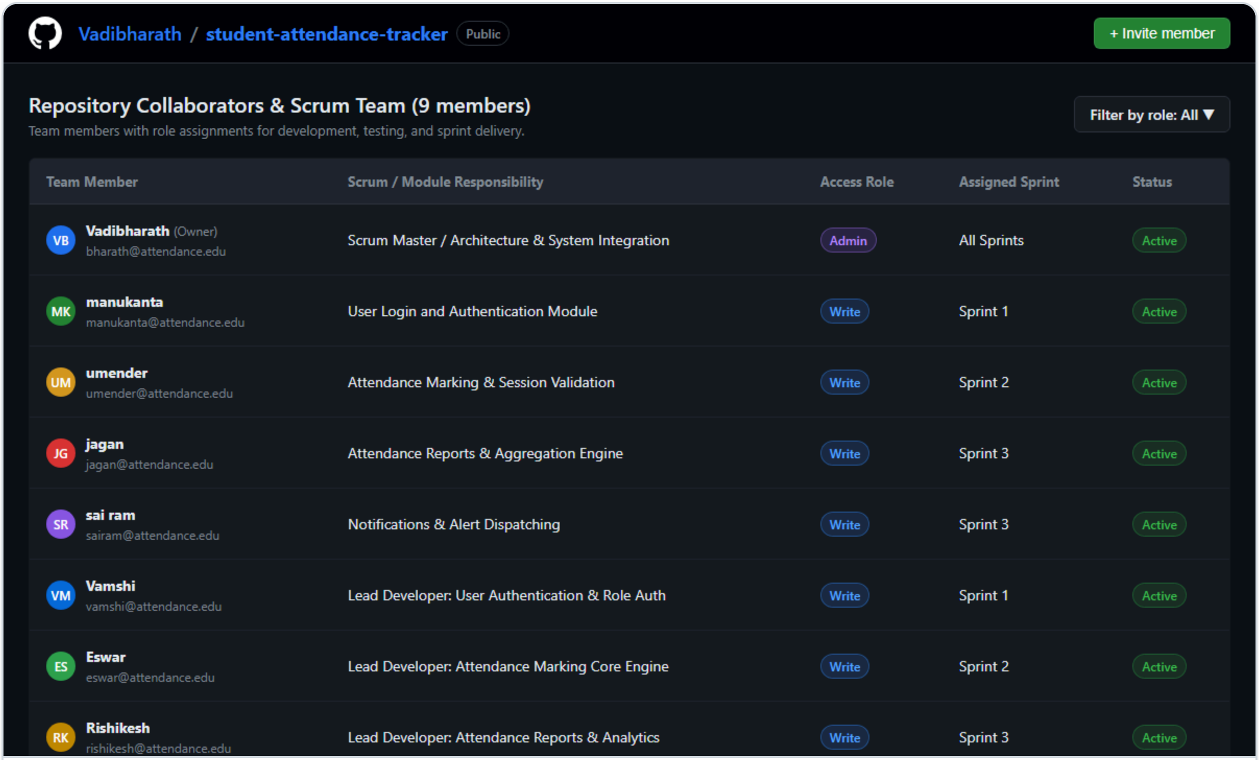
Manage collaborator permissions and team participation for Student Attendance Tracker.

Direct access
9 collaborators have write and triage access to this repository.
Base: Read & Write

Two-factor authentication (2FA) enforcement
All team members must have 2FA enabled to commit and merge PRs.
Enforced

Branch protection rules
main branch requires pull request review before merging & passing CI checks.
Protected (main)

Screenshot 1.1: Repository Collaboration and Access Management Settings



Repository Collaborators & Scrum Team (9 members)

Team members with role assignments for development, testing, and sprint delivery.

Team Member	Scrum / Module Responsibility	Access Role	Assigned Sprint	Status
VB **Vadibharath** (Owner) bharath@attendance.edu	Scrum Master / Architecture & System Integration	Admin	All Sprints	Active
MK **manukanta** manukanta@attendance.edu	User Login and Authentication Module	Write	Sprint 1	Active
UM **umender** umender@attendance.edu	Attendance Marking & Session Validation	Write	Sprint 2	Active
JG **jagan** jagan@attendance.edu	Attendance Reports & Aggregation Engine	Write	Sprint 3	Active
SR **sai ram** sairam@attendance.edu	Notifications & Alert Dispatching	Write	Sprint 3	Active
VM **Vamshi** vamshi@attendance.edu	Lead Developer: User Authentication & Role Auth	Write	Sprint 1	Active
ES **Eswar** eswar@attendance.edu	Lead Developer: Attendance Marking Core Engine	Write	Sprint 2	Active
RK **Rishikesh** rishikesh@attendance.edu	Lead Developer: Attendance Reports & Analytics	Write	Sprint 3	Active

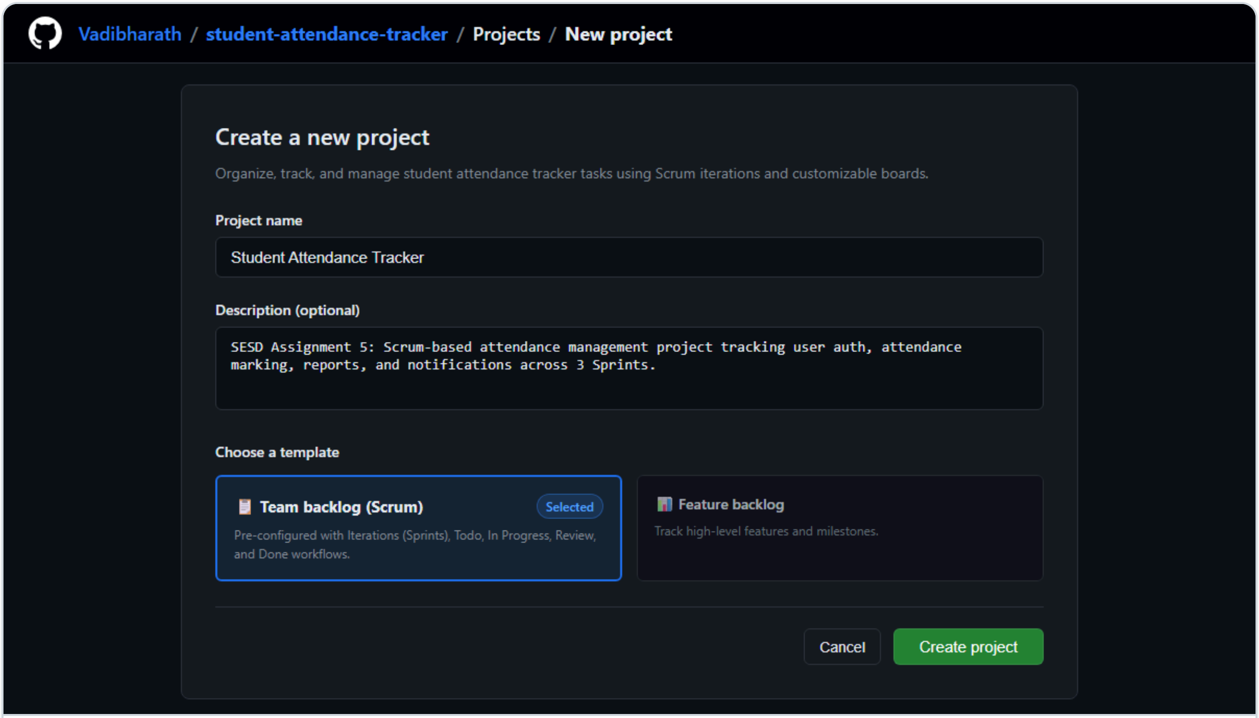
Screenshot 1.2: Repository Team Members & Scrum Module Assignments

Page 2: GitHub Project

Project Creation and Initial Board

Create Project

A GitHub Project is created for the Student Attendance Tracker. The repository issues are added to the project to organize the development tasks. Using the "Team Backlog" Scrum template, the team configures dedicated iterations (Sprints) and status views to support agile ceremonies.



Vadibharath / student-attendance-tracker / Projects / New project

Create a new project

Organize, track, and manage student attendance tracker tasks using Scrum iterations and customizable boards.

Project name

Student Attendance Tracker

Description (optional)

```
SESD Assignment 5: Scrum-based attendance management project tracking user auth, attendance
marking, reports, and notifications across 3 Sprints.
```

Choose a template

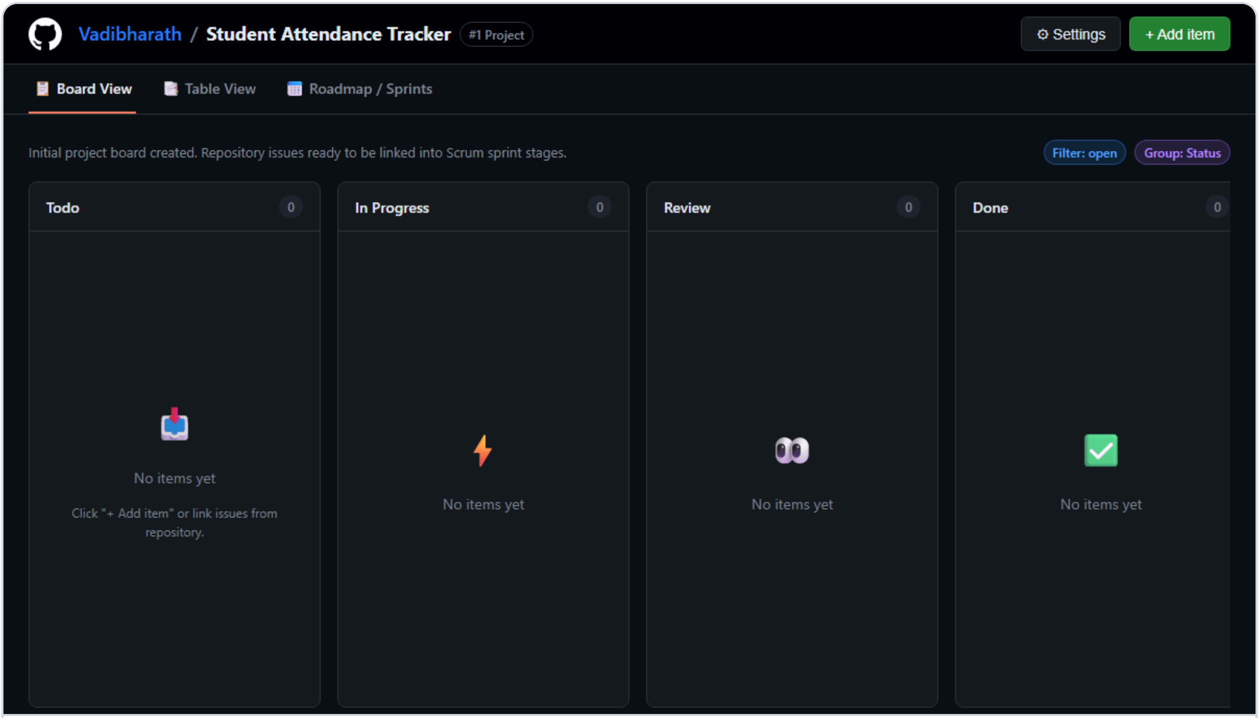
Team backlog (Scrum) Selected
Pre-configured with Iterations (Sprints), Todo, In Progress, Review, and Done workflows.

Feature backlog
Track high-level features and milestones.

Screenshot 2.1: Creating the GitHub Project with Team Backlog (Scrum) Template

Initial Board

The project board provides a central place to organize issues and track their progress through different stages. The initial board structure defines empty Kanban columns for **Todo**, **In Progress**, **Review**, and **Done**, ready for backlog decomposition and sprint allocation.


Vadibharath / Student Attendance Tracker
#1 Project
⚙ Settings
+ Add item
Board View
Table View
Roadmap / Sprints
Filter: open
Group: Status
Todo
0
In Progress
0
Review
0
Done
0

Initial project board created. Repository issues ready to be linked into Scrum sprint stages.

No items yet

Click "+ Add item" or link issues from repository.

No items yet

No items yet

No items yet

Screenshot 2.2: Initial Project Board with Scrum Stages (Todo, In Progress, Review, Done)

Page 3: Feature Issue and Backlog

Student Attendance Management

Student Attendance

The Student Attendance feature is created as a parent issue. It defines the requirements for recording, storing, and displaying student attendance. As the umbrella epic, it unites all institutional attendance requirements into clear engineering acceptance criteria.

Main Requirements

✓ Record attendance as Present or Absent.

✓ Store attendance date, subject/class code, and student details.

✓ Allow authorized faculty to mark attendance with credential verification.

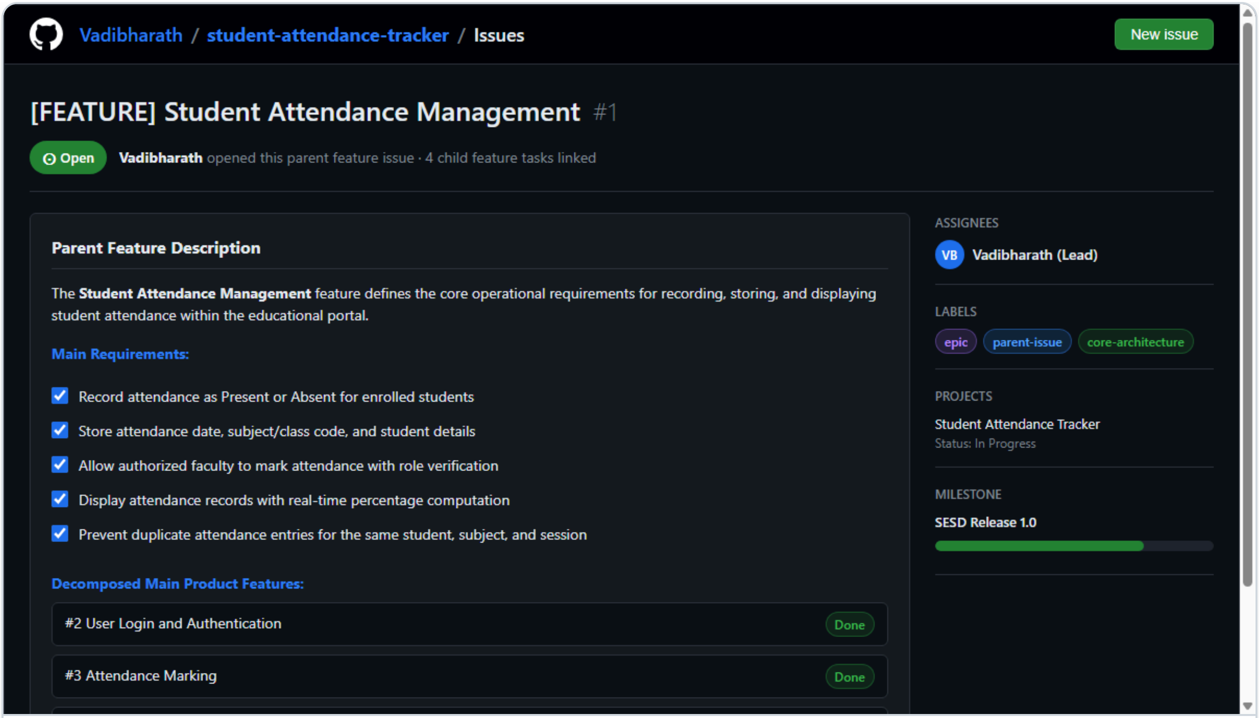
✓ Display attendance records and compute percentage in real time.

✓ Prevent duplicate attendance entries for the same student, subject, and session.

Feature List

The project board contains the following main feature issues:

1. **User Login and Authentication** (Issue #2)
2. **Attendance Marking** (Issue #3)
3. **Attendance Reports** (Issue #4)
4. **Notifications** (Issue #5)



[FEATURE] Student Attendance Management #1

Open **Vadibharath** opened this parent feature issue · 4 child feature tasks linked

Parent Feature Description

The **Student Attendance Management** feature defines the core operational requirements for recording, storing, and displaying student attendance within the educational portal.

Main Requirements:

- ☑ Record attendance as Present or Absent for enrolled students
- ☑ Store attendance date, subject/class code, and student details
- ☑ Allow authorized faculty to mark attendance with role verification
- ☑ Display attendance records with real-time percentage computation
- ☑ Prevent duplicate attendance entries for the same student, subject, and session

Decomposed Main Product Features:

- #2 User Login and Authentication — Done
- #3 Attendance Marking — Done

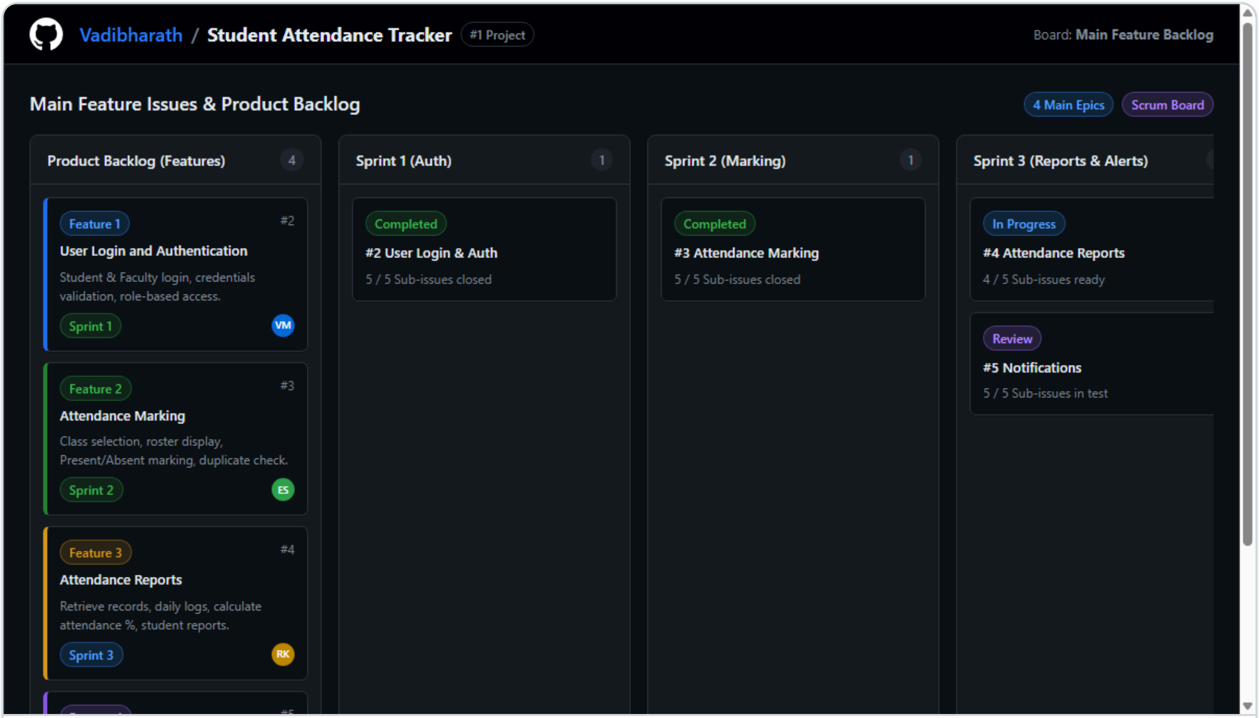
ASSIGNEES
VB Vadibharath (Lead)

LABELS
epic, parent-issue, core-architecture

PROJECTS
Student Attendance Tracker
Status: In Progress

MILESTONE
SESD Release 1.0

Screenshot 3.1: Parent Feature Issue #1 — Student Attendance Management


Vadibharath / Student Attendance Tracker #1 Project
Board: Main Feature Backlog
4 Main Epics
Scrum Board

Main Feature Issues & Product Backlog

Product Backlog (Features) 4

- Feature 1 #2 — **User Login and Authentication** — Student & Faculty login, credentials validation, role-based access. — Sprint 1 — VM
- Feature 2 #3 — **Attendance Marking** — Class selection, roster display, Present/Absent marking, duplicate check. — Sprint 2 — ES
- Feature 3 #4 — **Attendance Reports** — Retrieve records, daily logs, calculate attendance %, student reports. — Sprint 3 — RK

Sprint 1 (Auth) 1

- Completed — **#2 User Login & Auth** — 5 / 5 Sub-issues closed

Sprint 2 (Marking) 1

- Completed — **#3 Attendance Marking** — 5 / 5 Sub-issues closed

Sprint 3 (Reports & Alerts)

- In Progress — **#4 Attendance Reports** — 4 / 5 Sub-issues ready
- Review — **#5 Notifications** — 5 / 5 Sub-issues in test

Screenshot 3.2: Main Feature Issues Organized on the Project Board Backlog

Page 4: Product Backlog

Each main feature is divided into smaller Sub-issues so that the team can implement and track the work step by step. This granular breakdown provides actionable tickets for sprint planning and estimation.

1. User Login and Authentication

✓ Create login page.

✓ Add student and faculty login.

✓ Validate username and password.

✓ Implement role-based access.

✓ Test login functionality.

2. Attendance Marking

✓ Select class or subject.

✓ Display student list.

✓ Mark students Present or Absent.

✓ Save attendance records with date and session.

✓ Prevent duplicate entries for the same session.

✓ Test attendance marking.

3. Attendance Reports

✓ Retrieve attendance records from storage.

✓ Display daily attendance summaries.

✓ Calculate attendance percentage per student and subject.

✓ Generate student-wise reports.

✓ Test report generation.

4. Notifications

✓ Notify students about attendance updates.

✓ Send low-attendance alerts (<75% threshold).

✓ Notify students when attendance records are updated or corrected.

✓ Integrate notifications with attendance reports.

✓ Test notification functionality.

> **Scrum Parent-Child Completion Rule:**
> The parent issue becomes complete when its related Sub-issues are completed, tested, and reviewed via pull requests.

Page 5: Notifications Feature

Notifications and Alerts

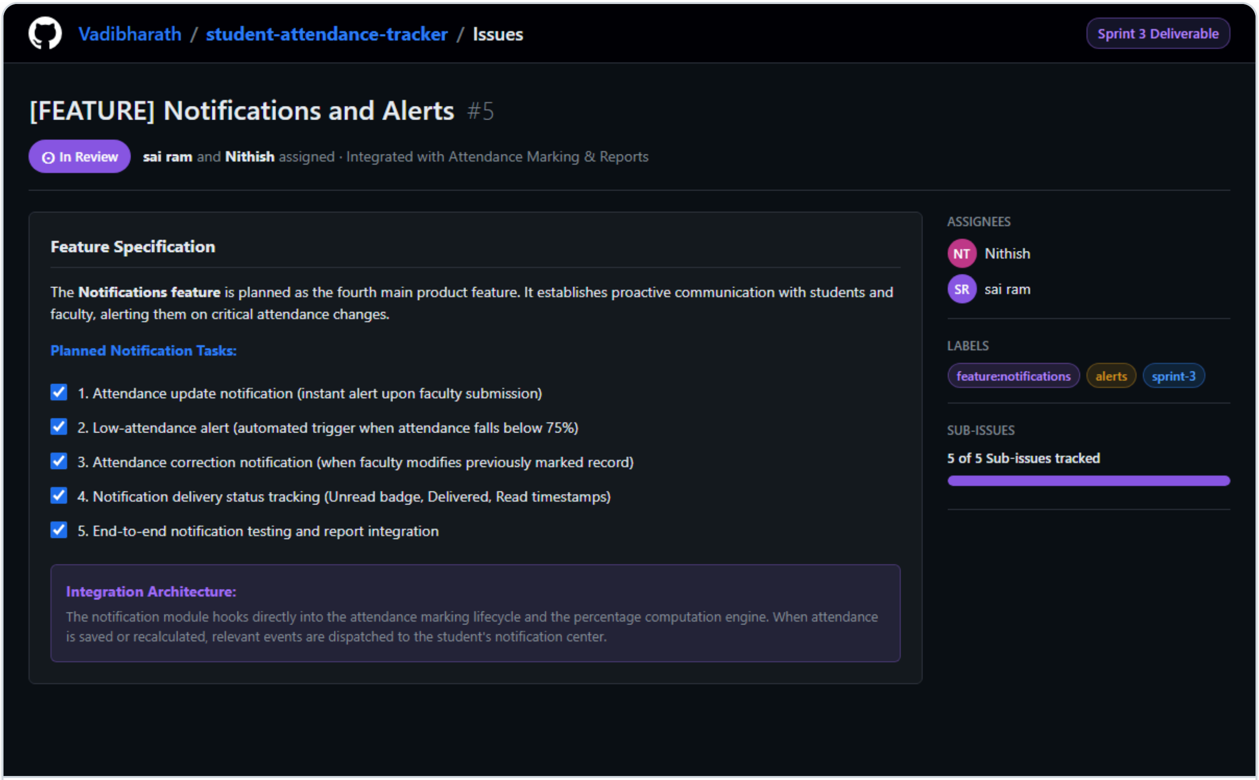
The Notifications feature is planned as the fourth main product feature. It provides an automated communication loop between faculty marking actions, attendance calculations, and student awareness.

Notification Tasks

1. **Attendance update notification:** Sent immediately to students when attendance for a lecture session is recorded.
2. **Low-attendance alert:** Automated warning banner and notification triggered when a student's attendance drops below 75%.
3. **Attendance correction notification:** Sent when faculty modifies or updates a past session entry.
4. **Notification delivery status:** Tracking unread badges, read status, and delivery timestamps.
5. **Notification testing:** Unit, integration, and UI acceptance testing of alerts.

Integration

The notification module is integrated with attendance marking and attendance reports so students can receive important updates in real time without polling or manual inquiries.



[FEATURE] Notifications and Alerts #5

In Review **sai ram** and **Nithish** assigned · Integrated with Attendance Marking & Reports

Feature Specification

The **Notifications feature** is planned as the fourth main product feature. It establishes proactive communication with students and faculty, alerting them on critical attendance changes.

Planned Notification Tasks:

- ☑ 1. Attendance update notification (instant alert upon faculty submission)
- ☑ 2. Low-attendance alert (automated trigger when attendance falls below 75%)
- ☑ 3. Attendance correction notification (when faculty modifies previously marked record)
- ☑ 4. Notification delivery status tracking (Unread badge, Delivered, Read timestamps)
- ☑ 5. End-to-end notification testing and report integration

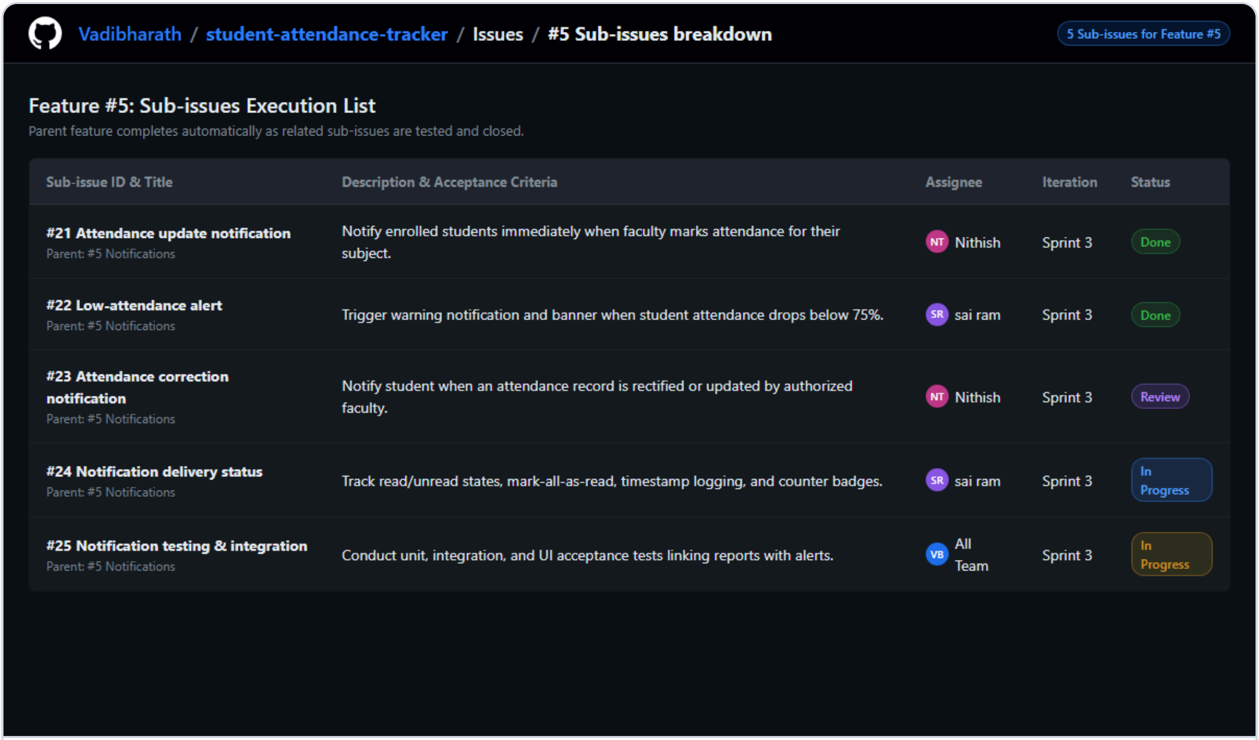
Integration Architecture:
The notification module hooks directly into the attendance marking lifecycle and the percentage computation engine. When attendance is saved or recalculated, relevant events are dispatched to the student's notification center.

ASSIGNEES
NT Nithish
SR sai ram

LABELS
feature:notifications alerts sprint-3

SUB-ISSUES
5 of 5 Sub-issues tracked

Screenshot 5.1: Notifications Parent Issue #5 Specification & Integration Scope


Vadibharath / student-attendance-tracker / Issues / #5 Sub-issues breakdown
5 Sub-issues for Feature #5

Feature #5: Sub-issues Execution List

Parent feature completes automatically as related sub-issues are tested and closed.

Sub-issue ID & Title	Description & Acceptance Criteria	Assignee	Iteration	Status
#21 Attendance update notification Parent: #5 Notifications	Notify enrolled students immediately when faculty marks attendance for their subject.	NT Nithish	Sprint 3	Done
#22 Low-attendance alert Parent: #5 Notifications	Trigger warning notification and banner when student attendance drops below 75%.	SR sai ram	Sprint 3	Done
#23 Attendance correction notification Parent: #5 Notifications	Notify student when an attendance record is rectified or updated by authorized faculty.	NT Nithish	Sprint 3	Review
#24 Notification delivery status Parent: #5 Notifications	Track read/unread states, mark-all-as-read, timestamp logging, and counter badges.	SR sai ram	Sprint 3	In Progress
#25 Notification testing & integration Parent: #5 Notifications	Conduct unit, integration, and UI acceptance tests linking reports with alerts.	VB All Team	Sprint 3	In Progress

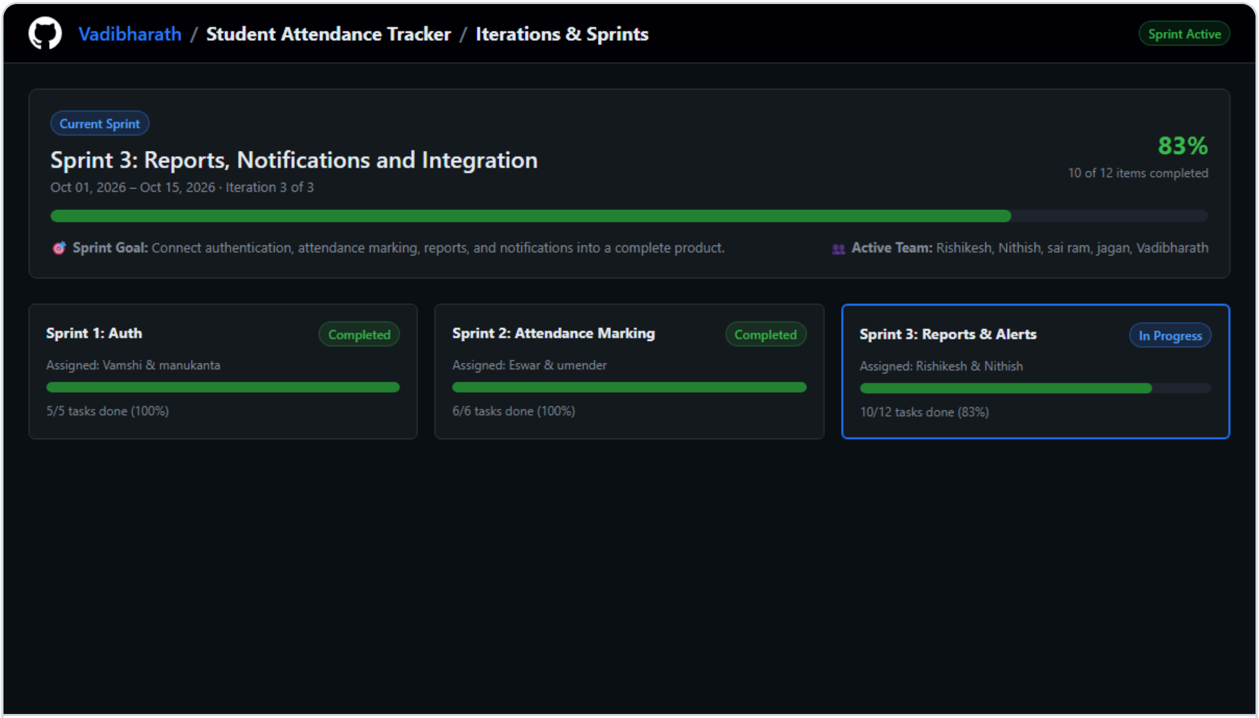
Screenshot 5.2: Decomposed Notification Sub-issues Linked to Parent Feature

Page 6: GitHub Project Board

Project Board and Current Iteration

Current Iteration

The current iteration groups tasks according to the team's sprint plan. Iteration 3 represents the integration phase where Attendance Reports and Notifications are connected into the core attendance engine.



Vadibharath / Student Attendance Tracker / Iterations & Sprints

Sprint Active

Current Sprint

Sprint 3: Reports, Notifications and Integration

Oct 01, 2026 – Oct 15, 2026 · Iteration 3 of 3

83%

10 of 12 items completed

Sprint Goal: Connect authentication, attendance marking, reports, and notifications into a complete product.

Active Team: Rishikesh, Nithish, sai ram, jagan, Vadibharath

Sprint 1: Auth Completed

Assigned: Vamshi & manukanta

5/5 tasks done (100%)

Sprint 2: Attendance Marking Completed

Assigned: Eswar & umender

6/6 tasks done (100%)

Sprint 3: Reports & Alerts In Progress

Assigned: Rishikesh & Nithish

10/12 tasks done (83%)

Screenshot 6.1: Current Iteration 3 Sprint Tracking & Velocity Progress Bar (83%)

Workflow Stages

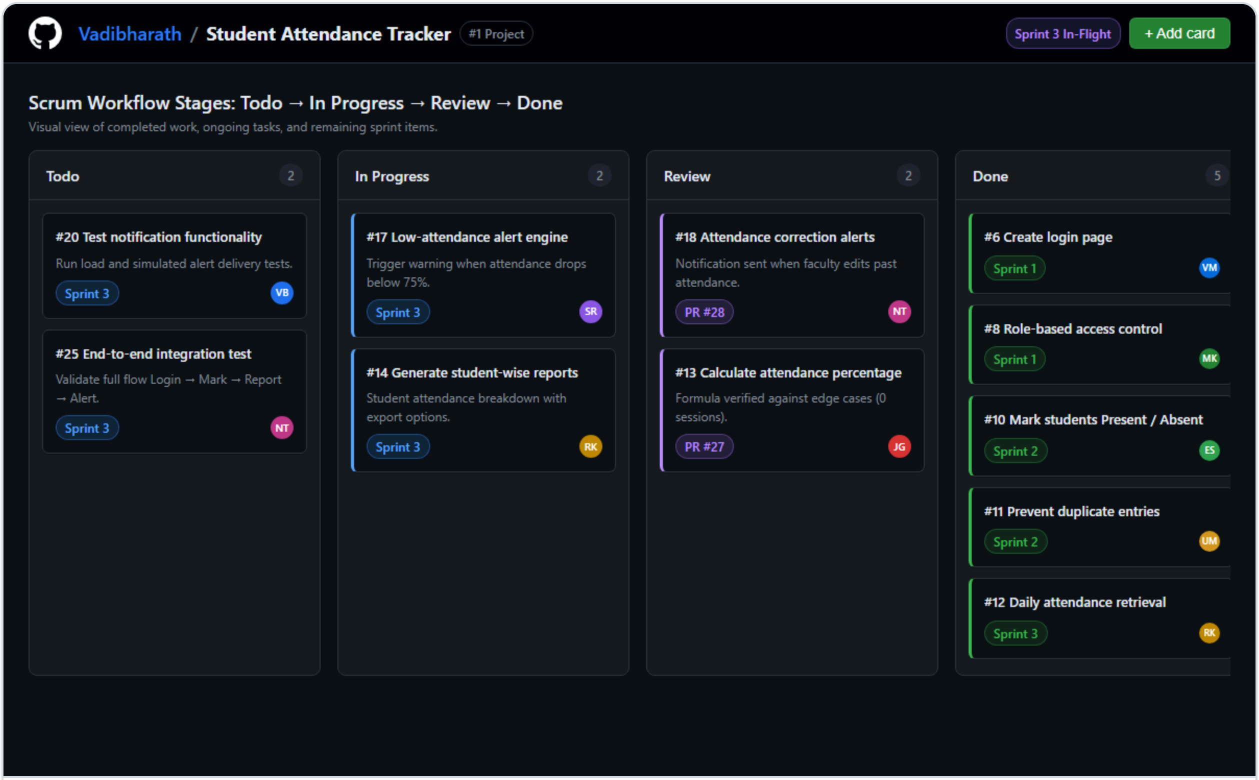
The team moves tasks across four distinct Scrum workflow columns:

- ✓ **Todo:** Backlog issues groomed and estimated for current sprint pickup.
- ✓ **In Progress:** Active development in local branches with assigned developers.
- ✓ **Review:** Pull requests submitted; peer code review and automated checks underway.

✓ **Done:** Code verified, merged to main branch, and verified against acceptance criteria.

Purpose

The project board provides a visual view of completed work, ongoing tasks, and remaining work. It promotes transparency and keeps all team members aligned with sprint velocity and blockers.



Scrum Workflow Stages: Todo → In Progress → Review → Done

Visual view of completed work, ongoing tasks, and remaining sprint items.

Todo 2

#20 Test notification functionality
Run load and simulated alert delivery tests.
Sprint 3 VB

#25 End-to-end integration test
Validate full flow Login → Mark → Report → Alert.
Sprint 3 NT

In Progress 2

#17 Low-attendance alert engine
Trigger warning when attendance drops below 75%.
Sprint 3 SR

#14 Generate student-wise reports
Student attendance breakdown with export options.
Sprint 3 RK

Review 2

#18 Attendance correction alerts
Notification sent when faculty edits past attendance.
PR #28 NT

#13 Calculate attendance percentage
Formula verified against edge cases (0 sessions).
PR #27 JG

Done 5

#6 Create login page
Sprint 1 VM

#8 Role-based access control
Sprint 1 MK

#10 Mark students Present / Absent
Sprint 2 ES

#11 Prevent duplicate entries
Sprint 2 UM

#12 Daily attendance retrieval
Sprint 3 RK

Screenshot 6.2: GitHub Project Board Displaying Issues Distributed Across Scrum Stages

Page 7: About the Project

Student Attendance Tracker

The Student Attendance Tracker is a Scrum-based software project designed to simplify attendance management in an educational institution. It allows faculty to mark attendance, students to view their attendance percentage, and the system to generate reports and notifications.

Team Responsibilities

Scrum Team Member	Primary Module Responsibility	Sprint Deliverable
manukanta	User Login and Authentication	Sprint 1 Lead
umender	Attendance Marking	Sprint 2 Lead
jagan	Attendance Reports	Sprint 3 Deliverable
sai ram	Notifications	Sprint 3 Deliverable

Product Goal

Login → Mark Attendance → View Attendance Reports → Receive Notifications

How the Team Is Developing It

Product Backlog → Sprint Planning → Sprint → Daily Scrum → Development → Sprint Review → Sprint Retrospective → Next Sprint

> **Scrum Rationale:**

> Using Scrum enables iterative delivery of user-visible increments at each sprint boundary, allowing the team to test, validate, and adapt features before full deployment.

Page 8: Sprint 1 — User Login and Authentication

> **Assigned Member:** Vamshi (with manukanta)

Sprint Tasks

1. **Design the login page:** Create a responsive, intuitive interface for students and faculty.
2. **Implement student and faculty login:** Support separate credential paths and roles.
3. **Validate login credentials:** Enforce format checks, error messaging, and password security.
4. **Add role-based access:** Route faculty to marking and analytics dashboards; route students to personal attendance and alerts.
5. **Test login functionality:** Execute automated unit tests and manual cross-device testing.

Sprint Goal

Allow authorized students and faculty to log in to the Student Attendance Tracker with secure, differentiated roles and protected session management.

Sprint 1 Output

A working authentication module that provides secure access to the attendance system. The module reliably identifies users, initializes session tokens, verifies institutional roles, and prevents unauthorized access to faculty functions.

Deliverable	Acceptance Criteria	Status
Login Interface	Responsive UI with role selection and input validation	**PASSED**
Role Authentication	Students restricted from faculty attendance modification	**PASSED**
Session State	User remains logged in across views with secure logout	**PASSED**

Page 9: Sprint 2 — Attendance Marking

> **Assigned Member:** Eswar (with umender)

Sprint Tasks

1. **Select class or subject:** Dropdown to filter active academic courses and sections.
2. **Display the student list:** Dynamic roster of enrolled students with roll numbers and names.
3. **Mark attendance as Present or Absent:** Interactive toggle switches and batch actions ("Mark All Present").
4. **Save attendance records with the date:** Persist record timestamp, subject code, faculty identifier, and roster status.
5. **Prevent duplicate entries:** Automated validation preventing duplicate attendance entries for the same student and session.
6. **Test attendance marking:** Verify boundary cases, offline persistence, and submission integrity.

Sprint Goal

Allow faculty to record student attendance accurately and efficiently with real-time roster rendering and duplicate entry safeguards.

Sprint 2 Output

A working attendance-marking module that stores attendance records for report generation. Faculty members can record morning/afternoon sessions in under 30 seconds with immediate persistence.

Deliverable	Acceptance Criteria	Status
Roster Selection	Dynamically populates student list based on selected course	**PASSED**
Batch & Individual Marking	Quick-toggle Present/Absent for entire section or individual students	**PASSED**

Deliverable	Acceptance Criteria	Status
Duplicate Protection	Warns and halts duplicate marking for identical subject + date + slot	**PASSED**

Page 10: Sprint 3 — Reports, Notifications and Integration

Attendance Reports — Rishikesh (with jagan)

- ✓ **Retrieve attendance records:** Efficient querying of historical records across dates and subjects.
- ✓ **Display daily attendance:** Daily summary breakdowns showing attendance counts and percentages.
- ✓ **Calculate attendance percentage:** Real-time formula `(Classes Attended / Total Held) × 100`.
- ✓ **Generate student-wise reports:** Individual progress cards highlighting attendance safety margin.
- ✓ **Test report generation:** Validation of zero-session edge cases and data export accuracy.

Notifications — Nithish and Entire Team (with sai ram)

- ✓ **Send attendance update notifications:** Trigger alert when faculty marks daily class.
- ✓ **Generate low-attendance alerts:** Proactive notification when cumulative attendance falls below 75%.
- ✓ **Notify students about attendance corrections:** Immediate notification upon record modifications.
- ✓ **Test notifications:** Delivery testing, timestamp validation, and mark-as-read functionality.
- ✓ **Integrate all modules:** Connect Authentication (Sprint 1), Marking (Sprint 2), and Reports & Alerts (Sprint 3).

Sprint Goal

Connect authentication, attendance marking, reports, and notifications to produce a complete Student Attendance Tracker.

Sprint 3 Output

An integrated system that allows users to log in, manage attendance, view reports, and receive notifications seamlessly.

Page 11: GitHub Workflow and Final Progress

GitHub Workflow

Issues → Sub-issues → Todo → In Progress → Review → Done

Scrum Activities

Daily Scrum: Team members discuss completed work, planned work, and blockers in short 15-minute syncs.

Sprint Review: The team demonstrates the features completed during the sprint to institutional stakeholders.

Sprint Retrospective: The team discusses what went well, what went wrong, and what can be improved for the next iteration.

Final Progress Progression

Sprint 1: User Login and Authentication

↓

Sprint 2: Attendance Marking

↓

Sprint 3: Attendance Reports + Notifications + Integration

↓

Final Student Attendance Tracker Application

Conclusion

The Student Attendance Tracker uses Scrum practices and GitHub collaboration tools to organize and monitor software development. GitHub Issues, Sub-issues, Iterations, Project

Board, commits, and pull requests help the team track tasks and work together efficiently. Through three focused sprints, the team transformed institutional requirements into a responsive, dependable, and user-centric web platform.